GLA UNIVERSITY, MATHURA

Assignment-I, 2026-2027

B.Tech. II Year, III/IV Semester

Subject Name- Probability and Statistics

Ques 1. Calculate the standard deviation and variance for the following table giving the age distribution of members of the house of commons.

Age	**10-20**	**20-30**	**30-40**	**40-50**	**50-60**	**60-70**
No. of members	**3**	**61**	**12**	**15**	**14**	**3**

Ques. 2. The mean of the following distribution is 35. Find the two missing frequencies f_1 and f_2. Given total frequency is 60.

Class Interval	**10–20**	**20–30**	**30–40**	**40–50**	**50–60**
Frequency	**5**	f_1	**20**	f_2	**5**

Ques. 3. A fair die is tossed twice. Let X be defined as the sum of two dice. Find the possible values for random variable "X" and its probability mass function P_X.

Ques. 4. The following data represent response times (seconds):

$$0, 0, 2, 2, 2, 3, 4, 5, 6, 7, 8, 9, 15, 18, 20, 25, 26, 27, 7, 7$$

a) Compute Q1, Q3 and IQR.
b) Identify mild and extreme outliers (if any).

Ques. 5. Determine the level of measurement in the examples given below: (Nominal, Ordinal, Interval, Ratio)

a. Time it takes to sing the National Anthem

b. Total annual income of a statistician

c. Body temperatures of bears in the North Pole

d. Rating an instructor as superior, above average, average, below average, or poor

Ques 6. Write a sample space for the following event: a die is rolled, and a coin is tossed.

Ques 7. Find the **mode** and the **median** for the following distribution:

Variable	**0-5**	**5-10**	**10-15**	**15-20**	**20-25**	**25-30**
frequency	**2**	**5**	**7**	**13**	**21**	**16**

Ques 8. A random variable **X** has the following probability function:

X	**-2**	**-1**	**0**	**1**	**2**	**3**
$P(X)$	**0.1**	**k**	**0.2**	**2k**	**0.3**	**k**

a) Find the value of k
b) Draw the graph of p.m.f.
c) Construct c.d.f F(x) and draw its graph.

Ques 9. The probability mass function $P(X = r)$of some discrete real-valued random variable $X$is given by the following table: (25 pts)

r	**0**	**1**	**2**	**3**	**4**
P(X = r)	0.35	0.15	?	0.10	0.20

(a) Fill in the missing value of $P(X = 2)$, explain your answer.

(b) Give the cumulative distribution function $F(X)$ using the following formula:

$$F(x) = P(X \leq x)$$

Ques 10. The probability distribution of a random variable X is given below. Find the expectation probability between 1 and 3 (inclusive).

X	**0**	**1**	**2**	**3**
P(X)	**0.1**	**0.3**	**0.4**	**0.2**